

The Off-Diagonal Bessel Kernel Resolves the Sobolev Restriction Unitarity Conjecture

Candidate full solution packet
arXiv:1607.01741, Remark 2.13

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Status. Candidate full affirmative solution, subject to expert review. If $\Omega \subset \mathbb{R}^n$ is nonempty and open and $\text{int}(\Omega^c) \neq \emptyset$, then

$$|_{\Omega} : \tilde{H}^s(\Omega) \longrightarrow H_0^s(\Omega) \quad \text{is unitary if and only if } s \in \mathbb{N}_0.$$

This proves the conjecture in Remark 2.13 of Hewett–Moiola [1].

1 Source conjecture

With the unitary Fourier transform, the source uses

$$(u, v)_{H^s(\mathbb{R}^n)} = \int_{\mathbb{R}^n} (1 + |\xi|^2)^s \widehat{u}(\xi) \overline{\widehat{v}(\xi)} d\xi. \quad (1)$$

Its Lemma 2.10 says that the restriction is unitary exactly when

$$\mathcal{D}(\Omega) \subset (H_{\Omega^c}^s)^{\perp}. \quad (2)$$

Proposition 2.12 proves the claimed classification when Ω is bounded. Remark 2.13 on printed page 6 asks whether it holds whenever Ω^c has nonempty interior:

Proof. The assumption that Ω^c is s -null means that $(H_{\Omega^c}^s)^\perp = (\{0\})^\perp = H^s(\mathbb{R}^n) \supset \mathcal{D}(\Omega)$, so part (iii) of Lemma 2.10 holds and hence the result follows. $\square$

Conversely, we can demonstrate that, when the complement of Ω is not negligible, $|\Omega : \tilde{H}^s(\Omega) \rightarrow H_0^s(\Omega)$ is not in general a unitary isomorphism except when $s \in \mathbb{N}_0$.

Proposition 2.12. *Assume that Ω is non-empty, open and bounded. Then the three equivalent statements in Lemma 2.10 hold if and only if s is a non-negative integer.*

Proof. We have seen in Lemma 2.2(iii) that $|\Omega : \tilde{H}^s(\Omega) \rightarrow H_0^s(\Omega)$ is a unitary isomorphism when $s \in \mathbb{N}_0$, for any Ω . When $s \notin \mathbb{N}_0$ and Ω is bounded we shall prove that this does not hold by showing that statement (iii) of Lemma 2.10 fails. Take any $\phi \in \mathcal{D}(\Omega)$ and define the translate $\phi_{\mathbf{d}}(\mathbf{x}) := \phi(\mathbf{x} - \mathbf{d})$ for $\mathbf{d} \in \mathbb{R}^n$. Then $\phi_{\mathbf{d}} \in \mathcal{D}(\mathbb{R}^n)$. In fact, since Ω is assumed bounded, for large enough $|\mathbf{d}|$ we have that $\phi_{\mathbf{d}} \in \mathcal{D}(\overline{\Omega}^c) \subset H_{\Omega^c}^s$, so that in particular $\text{supp } \phi \cap \text{supp } \phi_{\mathbf{d}}$ is empty. Define $\chi(\mathbf{d}) := (\phi, \phi_{\mathbf{d}})_{H^s(\mathbb{R}^n)} = (\phi(\cdot), \phi(\cdot - \mathbf{d}))_{H^s(\mathbb{R}^n)}$. Then the formula for the Fourier transform of a translate gives

$$\chi(\mathbf{d}) = (\phi, \phi_{\mathbf{d}})_{H^s(\mathbb{R}^n)} = \int_{\mathbb{R}^n} e^{i\mathbf{d} \cdot \boldsymbol{\xi}} \mu(\boldsymbol{\xi}) d\boldsymbol{\xi},$$

where $\mu(\boldsymbol{\xi}) := (1 + |\boldsymbol{\xi}|^2)^s |\hat{\phi}(\boldsymbol{\xi})|^2 = (1 + \xi_1^2 + \dots + \xi_n^2)^s |\hat{\phi}(\boldsymbol{\xi})|^2$, with $\boldsymbol{\xi} = (\xi_1, \dots, \xi_n)$. Since $\mu(\boldsymbol{\xi})$ is an element of $\mathcal{S}(\mathbb{R}^n)$, $\chi(\mathbf{d})$ is also an element of $\mathcal{S}(\mathbb{R}^n)$, with Fourier transform $\hat{\chi}(\boldsymbol{\xi}) = (2\pi)^{n/2} \mu(\boldsymbol{\xi})$. But for $s \neq 0, 1, 2, 3, \dots$ the function $\mu(\boldsymbol{\xi})$ does not extend to an entire function on $\mathbb{C}^n$ because the factor $(1 + \xi_1^2 + \dots + \xi_n^2)^s$ has singularities in $\mathbb{C}^n$. (E.g. for $n = 1$, these singularities occur at the points $\xi = \pm i$). Hence by the Paley–Wiener Theorem (see e.g. [4, Theorem 2.3.21]) $\chi(\mathbf{d})$ cannot have compact support in $\mathbb{R}^n$. As a result we can always find $\mathbf{d}$ with $|\mathbf{d}|$ large enough that $\phi_{\mathbf{d}} \in \mathcal{D}(\overline{\Omega}^c) \subset H_{\Omega^c}^s$ and $\chi(\mathbf{d}) = (\phi, \phi_{\mathbf{d}})_{H^s(\mathbb{R}^n)} \neq 0$. $\square$

Remark 2.13. *In proving the “only if” statement in Proposition 2.12 we required Ω to be bounded. With minor modifications the same proof works for some unbounded Ω . A first example is when Ω^c is bounded with non-empty interior. A second example is when either Ω itself or $\overline{\Omega}^c$, the interior of the complement of Ω , assumed to be non-empty, is contained in the hypograph $\{\mathbf{x} \in \mathbb{R}^n, x_n > g(x_1, \dots, x_{n-1})\}$, where $g : \mathbb{R}^{n-1} \rightarrow \mathbb{R}$ satisfies $\lim_{|\bar{\mathbf{x}}| \rightarrow \infty} g(\bar{\mathbf{x}}) = \infty$; the proof of Proposition 2.12 works in this case because $\chi(-\mathbf{d}) = \overline{\chi(\mathbf{d})}$. The result does not hold for every open set Ω , as Proposition 2.11 demonstrates. However, we conjecture that the statement of Proposition 2.12 holds for any Ω for which Ω^c has non-empty interior. But proving this conjecture appears to be an open problem.*

Remark 2.14. *Proposition 2.12 illustrates the fact that Sobolev norms with non-natural-number indices are non-local. In particular it implies that given any $s \in \mathbb{R} \setminus \mathbb{N}_0$, any $\phi \in \mathcal{D}(\mathbb{R}^n)$ and any (arbitrarily large) bounded set Ω containing the support of ϕ , there exists $\psi \in \mathcal{D}(\mathbb{R}^n)$ with support disjoint from Ω such that $(\phi, \psi)_{H^s(\mathbb{R}^n)} \neq 0$.*

2 Proof intuition

Integer Sobolev norms are local because their multiplier $(1 + |\xi|^2)^m$ is a polynomial; its inverse Fourier transform is supported at the origin. For every noninteger s , the opposite is true in the strongest possible form. Away from the origin, the distribution kernel of $(1 - \Delta)^s$ is a nonzero constant times a strictly positive function. This follows directly from the heat-kernel/Gamma integral.

Choose one nonnegative test function inside Ω and another inside $\text{int}(\Omega^c)$. Their supports are separated, so their H^s inner product is an ordinary double integral of that off-diagonal kernel. The integrand has one strict sign, and the inner product cannot vanish. Condition (2) therefore fails for every $s \notin \mathbb{N}_0$, without any geometric assumption beyond the existence of the two open regions.

3 The off-diagonal multiplier kernel

For $s \in \mathbb{C}$, let κ_s denote the distribution kernel with Fourier multiplier $(1 + |\xi|^2)^s$, normalized so that for separated test functions

$$(\phi, \psi)_{H^s(\mathbb{R}^n)} = \iint_{\mathbb{R}^n \times \mathbb{R}^n} \kappa_s(y - x) \phi(x) \overline{\psi(y)} dx dy. \quad (3)$$

Lemma 1 (Off-diagonal Bessel kernel). *For $x \neq 0$ and real s ,*

$$\kappa_s(x) = \frac{1}{(4\pi)^{n/2} \Gamma(-s)} \int_0^\infty t^{-s-n/2-1} \exp\left(-t - \frac{|x|^2}{4t}\right) dt. \quad (4)$$

Consequently $\kappa_m(x) = 0$ off the origin for $m \in \mathbb{N}_0$, while for every real $s \notin \mathbb{N}_0$ the function κ_s is smooth, nowhere zero, and has the constant sign of $1/\Gamma(-s)$ on $\mathbb{R}^n \setminus \{0\}$.

Proof. First suppose $\operatorname{Re} s < 0$. The Gamma identity gives

$$(1 + |\xi|^2)^s = \frac{1}{\Gamma(-s)} \int_0^\infty t^{-s-1} e^{-t} e^{-t|\xi|^2} dt.$$

The inverse Fourier kernel under the normalization in (3) is

$$(2\pi)^{-n} \int_{\mathbb{R}^n} e^{ix \cdot \xi} e^{-t|\xi|^2} d\xi = (4\pi t)^{-n/2} e^{-|x|^2/(4t)}.$$

Fubini's theorem yields (4) in this half-plane.

For fixed $x \neq 0$, the integral in (4) converges locally uniformly for every complex s : e^{-t} controls infinity and $e^{-|x|^2/(4t)}$ controls zero. Hence its product with the entire function $1/\Gamma(-s)$ is entire in s . On the other hand, $\mathcal{F}^{-1}((1 + |\xi|^2)^s)$, restricted to test functions supported away from the origin, is also an entire distribution-valued function of s ; on compact s -sets this follows from the polynomial growth of the multiplier and rapid decay of test-function Fourier transforms. Analytic continuation therefore proves (4) for all s off the origin.

For real s the integral is strictly positive. The reciprocal Gamma factor vanishes exactly when $s \in \mathbb{N}_0$ and is finite and nonzero otherwise. This proves every assertion. $\square$

4 Main theorem

Theorem 2. *Let $\Omega \subset \mathbb{R}^n$ be nonempty and open, with $\operatorname{int}(\Omega^c) \neq \emptyset$. Then*

$$|_\Omega : \tilde{H}^s(\Omega) \longrightarrow H_0^s(\Omega)$$

is a unitary isomorphism if and only if s is a nonnegative integer.

Proof. For $s = m \in \mathbb{N}_0$, the H^m inner product is local: by expanding $(1 + |\xi|^2)^m$ and applying Plancherel, it is a finite sum of L^2 inner products of matching derivatives. Hence test functions supported in Ω are orthogonal to H^m distributions supported in Ω^c . Condition (2), and therefore unitarity, follows. This is also Lemma 2.2(iii) of [1].

Now let $s \notin \mathbb{N}_0$. Choose nonempty balls U, V with

$$\overline{U} \subset \Omega, \quad \overline{V} \subset \operatorname{int}(\Omega^c),$$

and choose nonzero, nonnegative real functions $\phi \in \mathcal{D}(U)$ and $\psi \in \mathcal{D}(V)$. The compact supports are disjoint and have positive distance. Thus (3) is an ordinary absolutely convergent integral, and Lemma 1 gives

$$(\phi, \psi)_{H^s(\mathbb{R}^n)} = \iint_{U \times V} \kappa_s(y - x) \phi(x) \psi(y) dx dy \neq 0,$$

because the integrand has one strict sign wherever $\phi(x)\psi(y) > 0$. But $\psi \in \mathcal{D}(V) \subset H_{\Omega^c}^s$, so $\phi \notin (H_{\Omega^c}^s)^\perp$. Condition (2) fails, and the restriction cannot be unitary. $\square$

5 Verification, scope, and novelty

The decisive normalization check is $n = 1, s = -1$: formula (4) gives $\kappa_{-1}(x) = \frac{1}{2}e^{-|x|}$, agreeing with the source's explicit $H^{-1}(\mathbb{R})$ kernel. The proof uses only separated compact supports, so no singularity at the diagonal or distributional local term enters the double integral. The Gamma factor explains the sharp threshold: it vanishes precisely at nonnegative integers, exactly where the multiplier is a polynomial and the norm is local.

The theorem settles only the unitarity conjecture in Remark 2.13. It does not characterize surjectivity for rough open sets, nor the half-integer images mentioned elsewhere in the paper.

A bounded novelty search through 11 August 2026 checked the run registry and all lightweight indexes; the arXiv source and published DOI 10.1515/jaa-2017-0001; exact title, authors, proposition, conjecture, restriction-unitarity, nonempty-interior, Bessel-kernel, and citation searches. No later paper or independent resolution was found. Novelty confidence is moderate because bounded search is not a proof of priority.

6 Human-review recommendation

Recommended for expert review as a full solution. The main point to verify is the analytic continuation of the off-diagonal kernel in Lemma 1; after that lemma, the sign argument is immediate.

References

- [1] D. P. Hewett and A. Moiola, *A note on properties of the restriction operator on Sobolev spaces*, J. Appl. Anal. 23 (2017), 1–8; arXiv:1607.01741, Remark 2.13, p. 6.